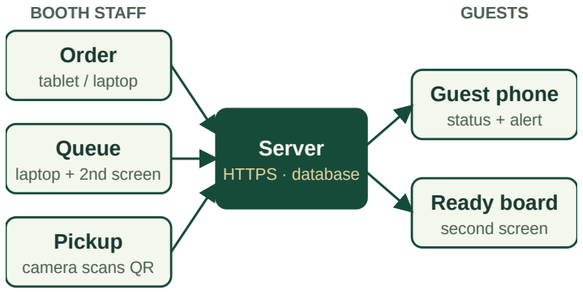
Beans&Beats — Technical Summary (Submission)

Christmas Together 2026 · 26–29 Nov · Hall F, Marina Bay Sands · Live: https://165-22-242-253.sslip.io · Ready board: /board · All profits to Singapore Red Cross Young Hearts

How it fits together



Each device saves taps first, then syncs when online.
Phones and the board update live in about 2 s; no app.

- **Order** on one tablet or laptop · **Queue** on a laptop (number keys) with a **second screen** for the Ready board · **Pickup** on any device with a camera
- Guests use **their own phone**: scan the QR, no app, no login
- One person can run it all on one device; more staff = more devices, kept in sync

Stack and architecture

Part	Built with
Staff app, guest page, board	React 19 + Vite, any browser
Works offline	Service worker + IndexedDB (Dexie)
Server	FastAPI, Python 3.14
Database	PostgreSQL 17
Live updates	Server-sent events; polling as backup
Hosting	Docker + Caddy HTTPS, DigitalOcean Singapore
Access	Barista / Booth Lead / Admin, checked by the server

- One live connection per page; pages re-check every 5 s (guest pages 20 s) if push drops
- Every change is saved with an audit entry in the same database step

Offline and sync

1. Every tap (order, Making, Ready, Collected, Cancel) is **saved on the device first**
2. When online, the device sends them **one at a time, in order**, to `POST /api/sync`
3. Each action has a **unique ID**: a resend is never counted twice
4. Status **only moves forward**; a manager's correction is never undone by an old offline tap
5. Each device has **its own letter** (A001, B001), so offline numbers never clash
6. Other devices, guest phones and the board update **within about 2 s** of the sync

- Keeps working through a Wi-Fi drop; reloading the page offline keeps every saved order
- Needs a connection: sign-in, booth settings, reports, CSV export, day reset

Personal data and PDPA

✓	**First name only**, used to call the guest and show on the board
✗	No phone, email, address or payment details
✗	No cookies, analytics or tracking on guest pages
✓	Guest page reachable only by its own **random link**
✓	Live updates sent to phones carry **no names**
✓	**Day reset clears that day's names**; sales and audit stay without names
✓	Only signed-in staff see names; the audit trail never stores them
✓	Payments stay on the organiser's POS
!	Daily backups (kept 14 days) may hold names from before a reset

Known limitations

- Devices sync through the internet; a booth-only network (no internet) is not built
- An offline device's changes reach others only after it reconnects
- Guest sound needs one tap first, and the page must stay open
- Staff devices and camera scanning need the HTTPS link
- Not yet tested at the hall: board reading at 3–5 m, every phone model, a 10-hour day
- Audit trail is locked in the app, not in the database itself
- Change the test logins before the event

The six core modules

Module	What it does
1 Order screen	Tap drink → Hot/Iced → Dairy/Oat → **Add to order** → first name → **Place order**. Number A001…, total to key into the POS, QR for the guest
2 Barista queue	Oldest first. **One tap Making, one tap Ready** (or type **012 + Enter**). Tabs: Active, To make, Received, Making, Ready, Late, Completed, Cancelled
3 Guest status page	Scan QR → order number, drinks, progress steps, orders ahead, **wait estimate**; chime, flash and title change when Ready
4 Ready board	Second screen / TV: **12 Ready + 12 In progress**, number beside name, pages every 8 s; Ready on the left or right
5 Menu controls	Change items and prices, **sold out** for a drink or oat milk; every device updates live
6 Daily sales report	Cups and totals **per day, by drink and option**; **CSV** to match the organiser's POS; cancelled, refunded and discarded counts

Extras

- **Pickup scan:** the guest's QR is the pickup ticket; the device camera reads it and marks Collected, with **Undo** for 10 s (or a Hand over tap)
- **Waiting-time estimate:** "Usually ready in about 7–11 min" from today's real pace; on/off for the board and for phones
- **Charity counter** panel on the board: cups today or all days, optional sales amount
- **Late drinks:** orange after 30 min, red after 60, guest warned; **Discard** button
- **Cancel** with a reason and refund status; **Cancelled** tab; Cancel in the popup and/or on the ticket
- **Booth settings (Booth Lead):** skip Making, collect on scan, reminder times, board side
- **Audit trail** (Admin), settings-change notice on other devices, **personal staff logins**

Why it stands out

No app, no hardware	Any browser; the camera reads the QR; no printer or reader
Wi-Fi drops? Keeps going	Saved on the device, synced once
Luckin-style guest alert	Live status, real wait estimate, chime
Total matches the POS	Refunds recorded; report counts only paid cups
Handles the crowd	300 phones, 10 min: 95% under 10 ms, 0 errors
Any crew size	One device or many, in sync
Safe by default	Server-checked roles, rate-limited logins, HTTPS, daily backups

How a drink flows

1. **Order:** barista keys it in; guest pays at the POS and scans the QR
2. **Queue:** Start making → Mark ready
3. **Guest phone:** chimes; name appears on the Ready board
4. **Pickup:** scan the guest's QR → Collected, off the board
5. **End of day:** Booth Lead checks every device is Synced → **Reset day**: names cleared, report kept

Tested

Latest backend checks: 91 tests passed on both SQLite and PostgreSQL (7 November).

- All 20 browser tests passed (7 Nov), including Wi-Fi drop and recovery
- Load test: 300 guest phones for 10 minutes on a 1-CPU server, no errors
- Every acceptance test is logged with its evidence